APPENDIX D – UML Diagrams

This appendix presents the UML diagrams used to model the structure and behaviour of the Trivia Fiest application. These diagrams illustrate user interaction, system flow, data structure, and component interaction within the system.

Figure D1: Use Case Diagram illustrating user interactions with the system

This diagram shows how users interact with the application, including selecting quiz options, answering questions, and viewing results. Additional features such as saving scores, exporting results, and uploading custom quizzes are also represented.

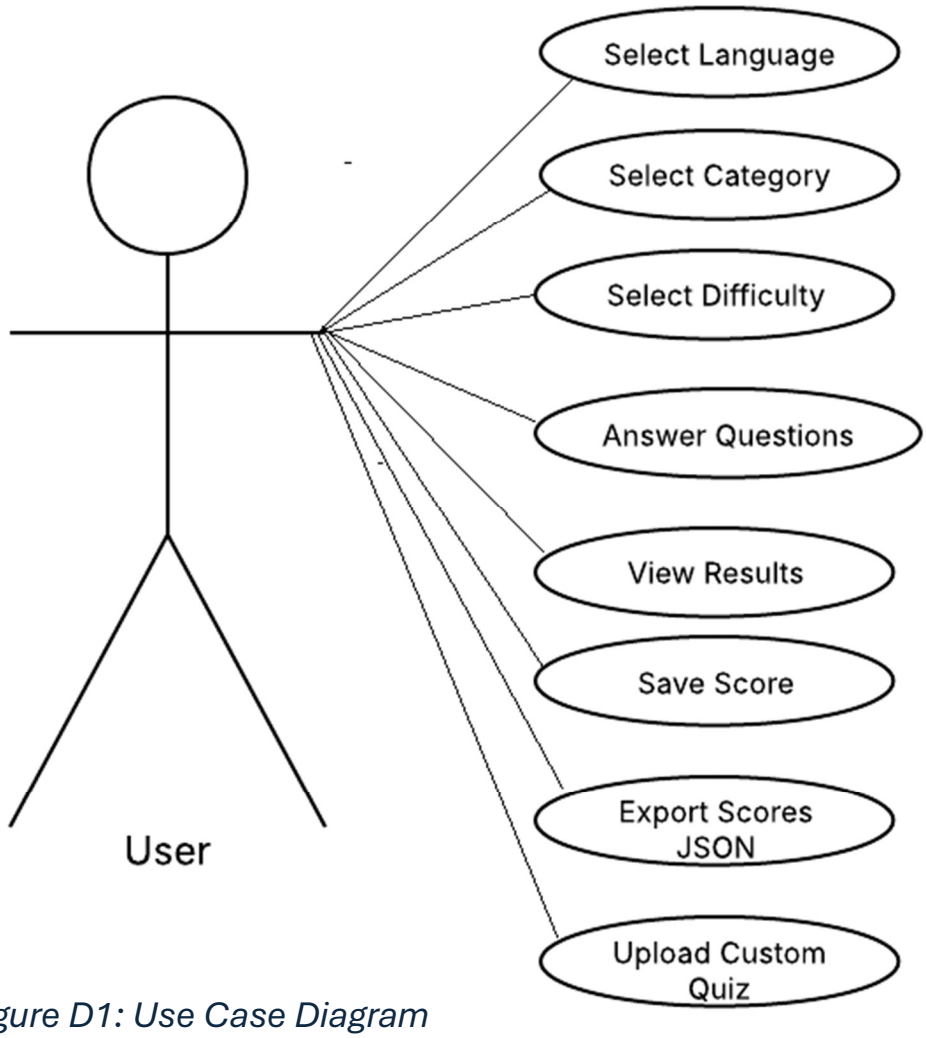

Figure D1: Use Case Diagram

FIGURE D2: Activity Diagram showing the overall quiz flow

This diagram illustrates the sequence of actions within the quiz, from initial setup to question answering and final results. It includes decision points such as timer expiration and question progression.

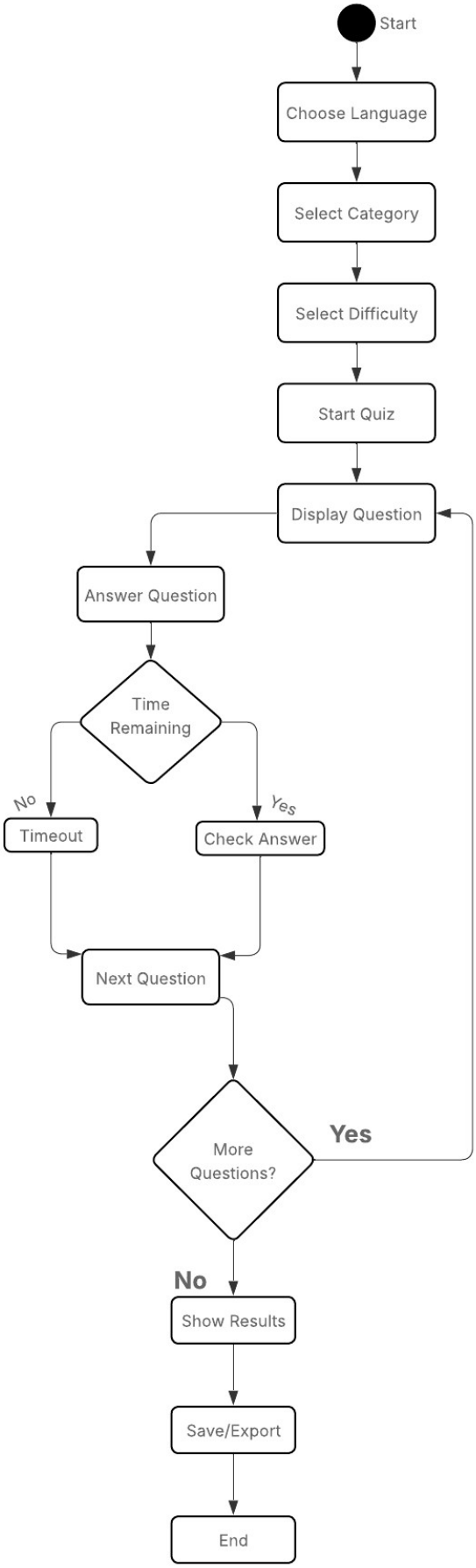

Figure D2: Activity Diagram

FIGURE D3 : Sequence Diagram

This diagram demonstrates how user input is processed by the system, showing the interaction between the user interface, application state, and DOM updates during question answering.

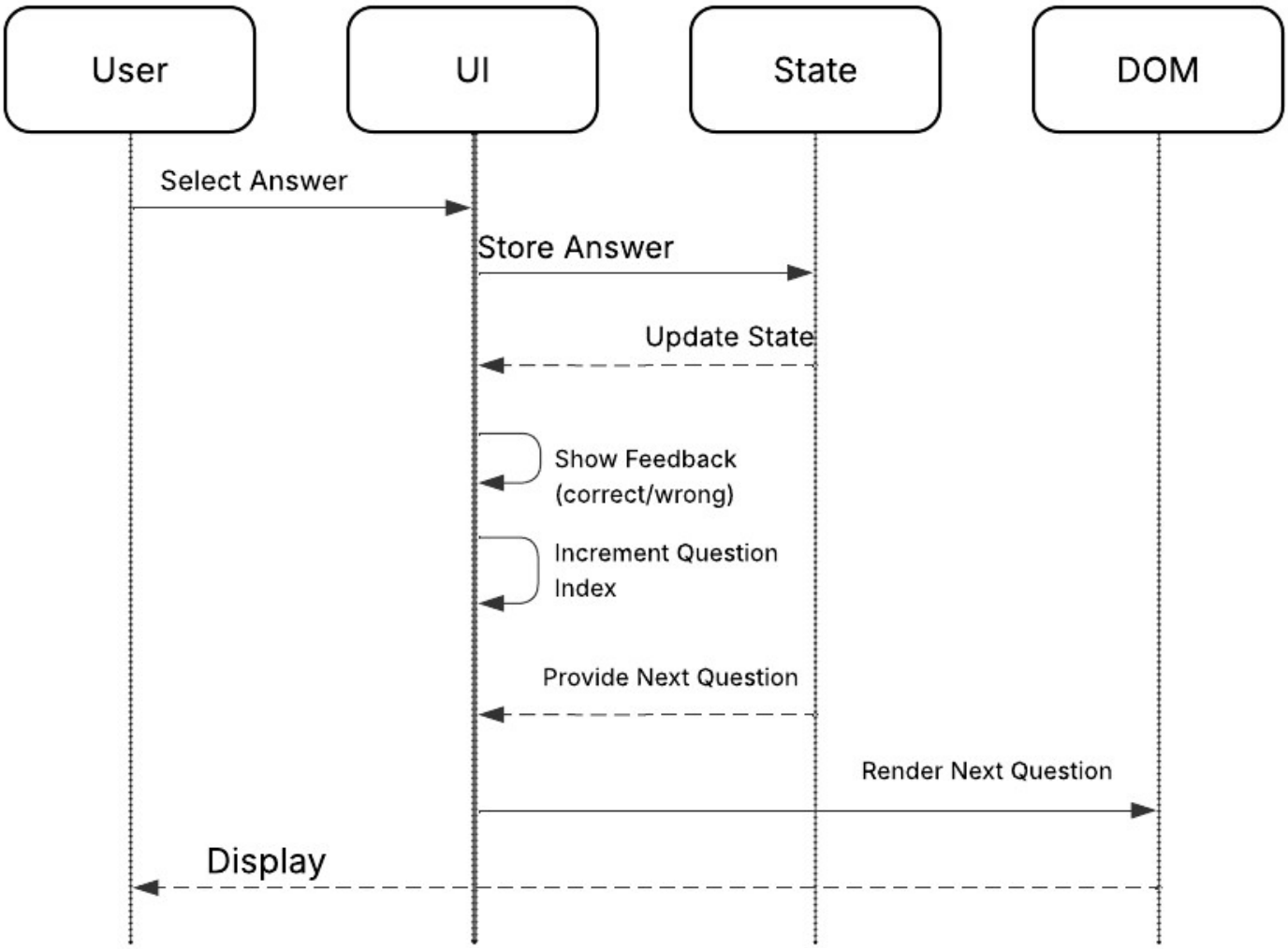

Figure D3: Sequence Diagram

FIGURE D4: Class Diagram

This diagram represents the structure of the application's data model, highlighting the centralised state object and its relationship with question data.

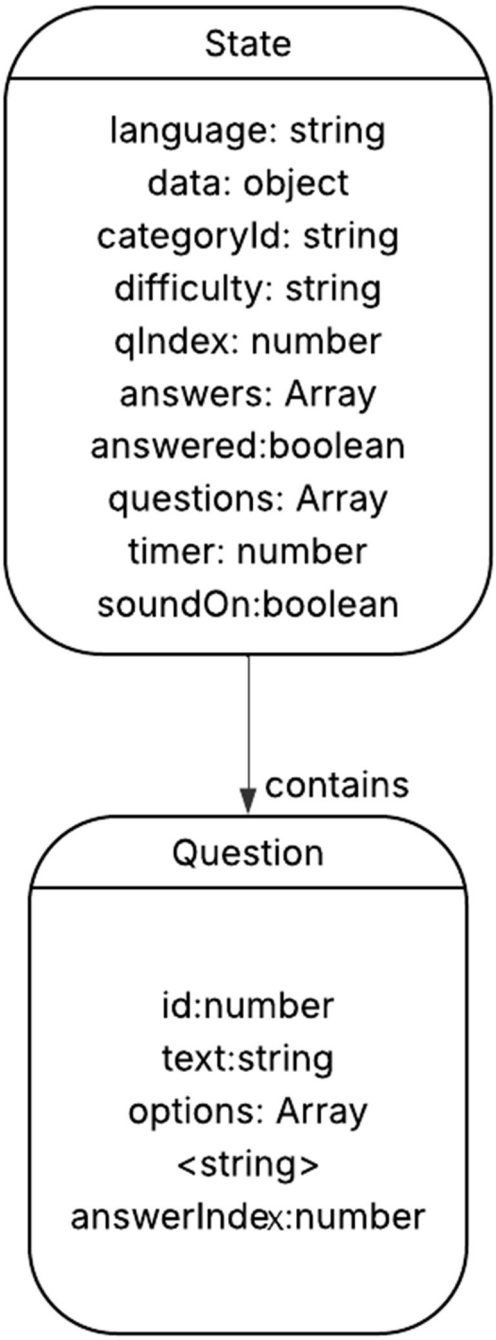

Figure D4: Class Diagram